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ÚSTAV POČÍTAČOVÝCH SYSTÉMŮ

3D NEURAL CELLULAR AUTOMATA

3D NEURÁLNÍ CELULÁRNÍ AUTOMATY

BACHELOR'S THESIS
BAKALÁŘSKÁ PRÁCE

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Abstract

Do tohoto odstavce bude zapsán výtah (abstrakt) práce v anglickém jazyce.

Abstrakt

Do tohoto odstavce bude zapsán výtah (abstrakt) práce v českém (slovenském) jazyce.

Keywords

Sem budou zapsána jednotlivá klíčová slova v anglickém jazyce, oddělená čárkami.

Klíčová slova

Sem budou zapsána jednotlivá klíčová slova v českém (slovenském) jazyce, oddělená čárkami.

Reference

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3D Neural Cellular Automata

Declaration

I hereby declare that this Bachelor's thesis was prepared as an original work by the author under the supervision of Ing. Karel Fritz I have listed all the literary sources, publications and other sources, which were used during the preparation of this thesis. I have used ChatGPT to correct spelling and other language mistakes. I used Grok when working on the software.

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Michal Repčík
November 25, 2025

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Chapter 3

Theoretical Foundations

3.1 Prerequisites

This thesis assumes the reader is familiar with basic concepts of deep learning, in particular convolutional neural networks, gradient-based optimisation, and automatic differentiation in PyTorch [11, 2].

A comprehensive introduction to these topics can be found in Goodfellow et al. [5] and in the official PyTorch documentation [2].

The rest of the thesis is self-contained with respect to cellular automata and their neural extensions.

3.2 Cellular Automata

Cellular automata (CA) are discrete computational systems that consist of a regular grid of cells, each of which can take on a finite number of states. The state of each cell evolves in parallel over discrete time steps according to a local update rule, which depends on the states of its neighboring cells [8].

Formally, a cellular automaton is defined by:

- A **lattice** of cells, typically 1D, 2D, or 3D.
- A **finite set of states** $\mathcal{S}$ that each cell can occupy.
- A **neighborhood function** $\mathcal{N}$ that defines which cells influence a given cell.
- An **update rule** $f : \mathcal{S}^{|\mathcal{N}|} \to \mathcal{S}$ that computes the next state [6].

CA were pioneered by von Neumann in the 1940s for self-replication studies [10], and later popularized by Wolfram's classification [13, 14] into four classes based on behavior:

- **Class 1 - Uniform**: quickly reaches a single state.
- **Class 2 - Periodic**: static or oscillating patterns.
- **Class 3 - Chaotic**: unpredictable, noisy evolution.
- **Class 4 - Complex**: long-lived structures and computation.

3.2.1 One-Dimensional Cellular Automata

The simplest non-trivial cellular automata live on a 1D line of cells. Each cell looks only at itself and its two immediate neighbours (radius = 1, neighbourhood size = 3). With two states (0 or 1), there are exactly 256 possible elementary rules, numbered from 0 to 255 in Wolfram's convention [13].

A rule number is a binary representation of the next-state lookup table. For example:

- Rule 90 produces the Sierpiński triangle from a single live cell (Class 3).
- Rule 30 creates chaotic, triangle-filled patterns that look random and are often used as pseudorandom generators.
- Rule 110 (Class 4) is the smallest known universal 1D CA - it supports gliders and glider collisions that can perform any computation [3].

3.2.2 Two-Dimensional Cellular Automata

In two dimensions the most common neighbourhood is the 3x3 Moore neighbourhood (8 surrounding cells + centre). With two states this gives $2^9 = 512$ possible inputs, so rules are usually described by birth/survival conditions instead of raw numbers.

The most famous example is Conway's Game of Life introduced in 1970 [4]. A cell becomes or stays alive if it has exactly 3 live neighbours (birth) or 2-3 live neighbours (survival); otherwise it dies. Despite these extremely simple rules, Game of Life exhibits:

- still lifes (e.g. block, beehive),
- oscillators (e.g. blinker, toad),
- spaceships (e.g. glider, lightweight spaceship),
- glider guns and other complex constructions.

Game of Life is Turing-complete [1] and has inspired almost all later work on self-organising and growing neural cellular automata, including the 2D Growing CA by Mordvintsev [9] that this thesis builds upon and extends into three dimensions.

3.2.3 Three-Dimensional Cellular Automata

In three dimensions, the Moore neighbourhood includes up to 26 adjacent cells (face-, edge-, and corner-adjacent) in a 3×3×3 cube. With binary states, this creates $2^{27} \approx 134$ million possible neighbourhood configurations, so practical 3D CA typically use totalistic or outer-totalistic rules that depend only on the sum of live neighbours (not their exact positions) [14].

Notable examples include:

- Von Neumann's 29-state self-replicating automaton (1950s), the foundational model for CA self-reproduction [10].
- Brian's Brain (1990s, 3-state rule): creates expanding „fireworks“ patterns via refractory periods [7].

- 3D Life variants (e.g. B3/S45678 or B5678/S45678): produce branching, crystal-like growths.

3D CAs have been studied since the 1950s but gained traction in the 2000s with better hardware for simulation [12]. Neural CA build on this by learning compact update rules that scale efficiently on GPUs, enabling stable 3D self-organisation.

3.3 Continuous and Differentiable Cellular Automata

3.4 Neural Cellular Automata

TODO

Chapter 4

Conclusion

TODO

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